MIL-D-43568(GL)

21 February 1968

MILITARY SPECIFICATION

DIPPER, STEEL, VITREOUS ENAMELED

1. SCOPE

1.1 This specification covers one type and size of enameled dipper.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-698 - Steel, Sheet and Strip, Low-carbon.
PPP-B-636 - Box, Fiberboard.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.

DRAWING

ARMY NATICK LABORATORIES

5-11-981 - Dipper, (Steel, Vitreous Enameled)

(Miniature copy of Drawing 5-11-981, identified as figure 1, is attached for information purposes only.)

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

FSC 65GP

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2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

Method C 282-65T - Tentative Method of Test for Acid Resistance of Porcelain Enamels.

Method C 286-66 - Terms Relating to Porcelain Enamel.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Preproduction sample.- When specified (see 6.2), before production is commenced, a sample item covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured by the supplier in the same facilities to be used for the manufacture of the production items.

3.2 Standard product.- The dipper delivered under this specification shall be the manufacturer's commercially current standard product, except for any changes necessary to comply with specification requirements.

3.3 Materials.- Materials shall conform to the applicable specifications, standards and requirements specified herein.

3.3.1 Steel, carbon.- The carbon steel sheets shall conform to cold rolled commercial quality (CRCQ) of QQ-S-698.

3.4 Construction.- The dipper shall consist of a body and handle and shall conform to the requirements specified herein and to the dimensions shown on Drawing No. 5-11-981.

3.4.1 Body.- The body shall be formed without seams from carbon steel specified in 3.3.1, having a minimum thickness of 0.0120 inch. The top edge of the body shall have a half round open bead.

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3.4.2 Handle.- The handle shall be fabricated from carbon steel specified in 3.3.1 and shall have one butting seam. The handle shall be conical in shape and shall be hollow with a slotted opening at the bottom side of the small diameter end. The handle shall be welded to the body at the small diameter end and where the periphery of the handle meets the half round open bead. A hole shall be provided at the end of the handle. The welds shall not break and the handle shall not be deformed when weight tested in accordance with 4.4.3.

3.5 Welding.- All surfaces to be welded shall be clean. Welding shall be accomplished to prevent the occurrence of burn holes, fractures, undercuts, weld metal or bare metal cracks, or incomplete fusion. The welds shall be sound, smooth, and shall blend into the contour of the handle. All scale or flux deposits shall be removed from the finished welds.

3.6 Finish.- The dipper shall have a white porcelain (vitreous) enamel finish. The porcelain (vitreous) enamel finish shall meet the class A or better acid resistance requirement of ASTM method C 282-65T when tested as specified in 4.4.2. The porcelain (vitreous) finish shall not be thicker than 12 mils, when tested as specified in 4.4.1. The porcelain (vitreous) enamel finish shall be glossy and free from discolorations, cracks, thin spots, or holes, and shall not show bare metal spots at corners. The handle and the rolled edge of the body may be furnished in blue, red, or black, high temperature, baked porcelain enamel finish.

3.7 Marking for identification.- The manufacturer's name or trademark of such known character, easily identifiable with said manufacturer shall be stamped, after coating, with an inorganic or ceramic ink in accordance with manufacturer's commercial practice or shall be applied on the dipper in accordance with MIL-STD-130. The surface opposite the markings shall show no signs of unevenness. Markings shall not cause distortion of any part of the dipper, where marked.

3.8 Workmanship.- Prior to the application of the porcelain (vitreous) enamel, the dipper shall be clean and shall not contain any burrs, slivers, or sharp edges; rough die, tool, gouge, or grind marks. Welding shall be accomplished in a manner that shall prevent the occurrence of pits and flashes. The dipper shall not be fractured, buckled, bent, punctured, chipped, malformed, or damaged.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Reproduction sample inspection.- When a preproduction sample is required, it shall be examined for defects listed in table I, for dimensions specified, and tested as specified in 4.4.1 and 4.4.2.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 Process examination.-

4.3.2.1 In process examination.- Examination of the dipper shall be made to establish conformance with the requirements specified herein. Whenever nonconformance is noted, correction shall be made to the affected items.

(a) 3.4.1 - Bowl formed without seams.
(b) 3.4.2 - Handle welded to body and to half round open bead.
(c) 3.8 - Clean prior to application of porcelain (vitreous) enamel.

4.3.2.2 In process testing.- Prior to the application of the vitreous enamel coating the handle shall be tested in accordance with 4.4.3. The inspection level shall be S-2 with an acceptable quality level (AQL) of 6.5 defects, expressed in terms of defects per hundred units. The lot shall consist of all dippers offered for inspection at one time. The sample unit shall be one dipper.

4.3.3 End product inspection.- The inspection lot shall be the number of dippers offered for inspection at one time. The sample unit for this inspection shall be one completely fabricated dipper.

4.3.3.1 Visual examination.- Examination shall be made of the dipper for defects in table I. The inspection level shall be level II with an AQL of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

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TABLE I.- Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not porcelain (vitreous) enamel finish	X	
Porcelain enamel (for definitions of defects consult ASTM C 286-66)	Not color specified		X
	Any discoloration, chips, fish scale, burn off, crazing beads, lumps, or scumming and blisters	X	
Construction and workmanship (applicable to all components and assemblies)	Any burr(s), slivers, or sharp edges; rough die, tool, gouge, or grind marks. Fractured, buckled, bent, punctured, chipped, malformed, or damaged	X	
Body	Not half round open bead on edge		X
Handle	Not a butting seam		X
	Not conical		X
	Hole in handle missing		X
Welding	Any burn holes, fractures, undercuts, weld or base metal cracks, or incomplete fusion	X	
	Weld not sound, smooth, or does not blend into contour of handle		X
Identification marking	Missing, incomplete, illegible, not proper location		X
	Surface opposite marking shows penetration or unevenness		X

4.3.3.2 Dimensional examination.- Inspection shall be made of the sample unit to determine compliance with dimensions specified. Any deviation from specified requirements shall constitute a defect. The inspection level shall be level S-2 and the AQL shall be 4.0 defects, expressed in terms of defects per hundred units.

4.3.3.3 Testing of the end item.- The end item shall be tested as specified in 4.4.1 and 4.4.2. The inspection level shall be S-2 with an AQL of 6.5 defects, expressed in terms of defects per hundred units.

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4.3.4 Examination of preparation for delivery.- An examination shall be made to determine that packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be scored in accordance with defects set forth in table II. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for delivery at one time. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

TABLE II.- Examination of preparation for delivery

Examine	Defect
Markings (exterior and interior container)	Missing; incorrect; illegible; of improper size, location sequence, or method of application.
Materials	Component missing, damaged, or otherwise defective
Contents (interior)	More or less than required.

4.3.4.1 Examination of shipping container.- When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing, and reinforcing shall be in accordance with that specification.

4.4 Tests.-

4.4.1 Porcelain (vitreous) enamel thickness test.- The thickness of the porcelain (vitreous) enamel finish shall be measured on the bottom flat surfaces, i.e. interior and exterior, of the dipper at three spots, away from edges, with a magnetic thickness gauge or other suitable means accurate to plus or minus 1 mil. The average thickness at three measured spots shall be used for determining thickness conformance to 3.6.

4.4.2 Acid-resistance test.- The porcelain (vitreous) enamel finish shall be tested in accordance with ASTM method C 282-65T to determine compliance with the acid resistance requirements specified in 3.6.

4.4.3 Handle test.- The dipper shall be loaded with weights equivalent to 1-1/4 times the weight of the water the dipper is intended to hold. The dipper shall then be lifted by the handle for 1 minute and examined to determine compliance with the deformation and weld requirements of 3.4.2.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

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5.1.1 Level A.- Twelve dippers shall be packaged in the most compact manner in a snug-fitting fiberboard box conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. The dippers shall be prevented from contacting each other by slip sheeting or wrapping with minimum 30 pound basis weight (24 X 36 - 500) kraft paper. Box closure shall be in accordance with the appendix of PPP-B-636.

5.1.2 Level C.- The dippers shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.- Seventy-two dippers, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V3s of PPP-B-636. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat strapping or tape banding in accordance with the appendix of PPP-B-636.

5.2.2 Level B.- Unless otherwise specified (see 5.2.2.1), seventy-two dippers, packaged as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. Each shipping container shall be closed in accordance with method II of the appendix of PPP-B-636.

5.2.2.1 When specified (see 6.2), the shipping container shall be V3c or V3s, fabricated and closed in accordance with PPP-B-636.

5.2.3 Level C.- Dippers, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The dipper covered by this specification is intended for the collection of mosquito larvae in water samples.

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When a preproduction sample is required (see 3.1).
(c) When V3c or V3s fiberboard is required for level B shipment (see 5.2.2.1).
(d) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Supersession.- This specification includes the requirements for the enameled dipper formerly contained in RR-E-516.

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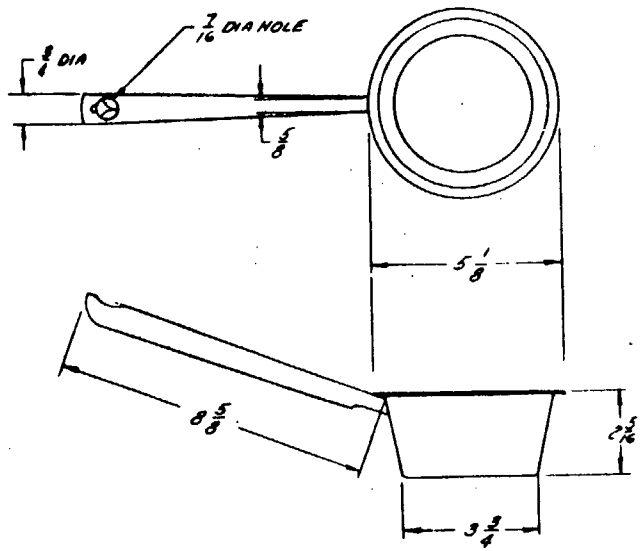


FIGURE 1 DIPPER
(STEEL, VITREOUS ENAMELED)

ITEM NO	DESCRIPTION	QTY REQD	MATERIAL	MATL SPEC	NO.

ITEM LIST

THIS FIGURE IS A MINIATURE COPY OF ARMY NATICK LABORATORIES DRAWING NUMBER 13/5123, REVISION NUMBER A, DATED 11-9-28. THIS COPY IS FOR INFORMATION ONLY; THE DRAWING REFERENCED IN PARAGRAPH 2.1 GOVERNS.

FIGURE – 1

DIPPER
(STEEL, VITREOUS ENAMELED)

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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